

-----CODE PARTS-----

```
Open 230104004174.c ~/Desktop
1#include <stdio.h>
2
3int main() {
4    /* Check for errors. */
5    FILE *dataFile;
6    dataFile = fopen("//home//gorkem//Desktop//data.txt" , "r");
7    if (dataFile == NULL) {
8        printf("An error occured while opening the file. \n");
9        return 1;
10    }
11
12    int sayi1, sayi2, sayi3;
13    int firstX_0,firstX_1= 0; /* these are flags */
14    int max_x_0 , max_x_1 , max_y_0 ,max_y_1 ,min_x_0 , min_x_1 , min_y_0 ,min_y_1;
15    while (fscanf(dataFile, "%d %d %d", &sayi1, &sayi2, &sayi3) != EOF) {
16
17        if(sayi3 == 0){ /* checking the points where class is 0. */
18            if(firstX_0 == 0){ /*since the first reading , I set the max and min values equal to the first number. */
19                max_x_0 = sayi1;
20                min_x_0 = sayi1;
21                max_y_0 = sayi2;
22                min_y_0 = sayi2;
23                firstX_0 = 1; /* I changed the flag to 1. so that it would sync only the first time. */
24            }
25            if(sayi1 > max_x_0){ /* check for max and min values.*/
26                max_x_0 = sayi1;
27            }
28            if(sayi2 > max_y_0){
29                max_y_0 = sayi2;
30            }
31            if(sayi1 < min_x_0){
32                min_x_0 = sayi1;
33            }
34            if(sayi2 < min_y_0){
35                min_y_0 = sayi2;
36            }
37
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Open 230104004174.c ~/Desktop
31            if(sayi1 < min_x_0){
32                min_x_0 = sayi1;
33            }
34            if(sayi2 < min_y_0){
35                min_y_0 = sayi2;
36            }
37
38        }else if(sayi3 == 1){ /* checking the points where class is 0. */
39            if(firstX_1 == 0){ /*since the first reading , I set the max and min values equal to the first number. */
40                max_x_1 = sayi1;
41                min_x_1 = sayi1;
42                max_y_1 = sayi2;
43                min_y_1 = sayi2;
44                firstX_1 = 1; /* I changed the flag to 1. so that it would sync only the first time. */
45            }
46            if(sayi1 > max_x_1){ /* check for max and min values.*/
47                max_x_1 = sayi1;
48            }
49            if(sayi2 > max_y_1){
50                max_y_1 = sayi2;
51            }
52            if(sayi1 < min_x_1){
53                min_x_1 = sayi1;
54            }
55            if(sayi2 < min_y_1){
56                min_y_1 = sayi2;
57            }
58        }
59    }
60
```

```
Open 230104004174.c ~/Desktop
58     }
59 }
60
61 /* QUESTION - 2*/
62 int sepx,sepy;
63 int r1, r2, r3, r4;
64 int errorCount = 0;
65 int minError , minErrorFlag = 0;
66 int bestSeparatorX, bestSeparatorY , bestR1,bestR2,bestR3,bestR4 ;
67 int minValueForSeperatorX , maxValueForSeperatorX , minValueForSeperatorY , maxValueForSeperatorY;
68
69 if(min_x_0 < min_x_1 ){ /* Check max and min values of x and y For separators.*/
70     minValueForSeperatorX = min_x_0;
71 }else{
72     minValueForSeperatorX = min_x_1;
73 }
74 if(min_y_0 < min_y_1 ){
75     minValueForSeperatorY = min_y_0;
76 }else{
77     minValueForSeperatorY = min_y_1;
78 }
79
80 if(max_x_0 < min_y_1 ){
81     maxValueForSeperatorX = max_x_0;
82 }else{
83     maxValueForSeperatorX = min_y_1;
84 }
85 if(max_y_0 < max_y_1 ){
86     maxValueForSeperatorY = max_y_0;
87 }else{
88     maxValueForSeperatorY = max_y_1;
89 }
90
```

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Open 230104004174.c ~/Desktop
88     maxValueForSeperatorY = max_y_1;
89 }
90
91 for (sepx = minValueForSeperatorX; sepx <= maxValueForSeperatorY; sepx++) { /* calculate every possibility for separators and labels.*/
92     for (sepy = min_x_0; sepy <= maxValueForSeperatorY; sepy++) {
93         for (r1 = 0; r1 <= 1; r1++) {
94             for (r2 = 0; r2 <= 1; r2++) {
95                 for (r3 = 0; r3 <= 1; r3++) {
96                     for (r4 = 0; r4 <= 1; r4++) {
97                         /* reverting to the beginning of the file for reading.*/
98                         FILE *dataFile2;
99                         dataFile2 = fopen("/home/gorken/Desktop/data.txt" , "r");
100
101                         errorCount = 0;
102
103                         while (fscanf(dataFile2, "%d %d %d", &say1, &say12, &say13) != EOF) { /* check one by one and take the best combination */
104                             if (say11 <= sepx && say12 > sepy) { /*with the least errorCount. */
105                                 if (say13 != r1) {
106                                     errorCount += 1;
107                                 }
108                             } else if (say11 > sepx && say12 > sepy) {
109                                 if (say13 != r2) {
110                                     errorCount += 1;
111                                 }
112                             } else if (say11 <= sepx && say12 <= sepy) {
113                                 if (say13 != r3) {
114                                     errorCount += 1;
115                                 }
116                             } else if (say11 > sepx && say12 <= sepy) {
117                                 if (say13 != r4) {
118                                     errorCount += 1;
119                                 }
120                             }
121                         }
122                         if(minErrorFlag == 0){ /* I set the first ErrorCount equal to minError. */
123                             minError = errorCount;
124                             minErrorFlag = 1; /* I set the minErrorFlag to 1. for prevent it from looping again */
125                         }else{
126                             if(errorCount < minError){ /*take the best combination with for errorcount. */
127                                 minError = errorCount;
128                                 bestSeparatorX = sepx;
129                                 bestSeparatorY = sepy;
130                                 bestR1 = r1;
131                                 bestR2 = r2;
132                                 bestR3 = r3;
133                                 bestR4 = r4;
134                             }
135                         }
136                     }
137                 }
138             }
139         }
140     }
141 }
```

```
Open 230104004174.c ~/Desktop
132         bestR3 = r3;
133         bestR4 = r4;
134     }
135 }
136 }
137 fclose(dataFile2); /* due to revert beginning.*/
138 }
139 }
140 }
141 }
142 }
143 }
144 }
145 fclose(dataFile); // close the file.
146
147 FILE *dataFile3;
148 dataFile3 = fopen("//home//gorkem//Desktop//test 1.txt" , "r");
149 if (dataFile3 == NULL) {
150     printf("An error occured while opening the file. \n");
151     return 1;
152 }
153 int trueCountFor2 = 0;
154 int numberOfDataPoints = 0;
155 while (fscanf(dataFile3, "%d %d %d", &sayi1, &sayi2, &sayi3) != EOF){
156     numberOfDataPoints += 1;
157     if (sayi1 <= bestSeparatorX && sayi2 > bestSeparatorY) { /* check the points according to best Separators*/
158         if (sayi3 == bestR1) {
159             trueCountFor2 += 1; /* Increase trueCountFor2 by 1 for each correct match.*/
160         }
161     } else if (sayi1 > bestSeparatorX && sayi2 > bestSeparatorY) {
162         if (sayi3 == bestR2) {
163             trueCountFor2 += 1;
164         }
165     } else if (sayi1 <= bestSeparatorX && sayi2 <= bestSeparatorY) {
166         if (sayi3 == bestR3) {
167             trueCountFor2 += 1;
168         }
169     } else if (sayi1 > bestSeparatorX && sayi2 <= bestSeparatorY) {
170         if (sayi3 == bestR4) {
171             trueCountFor2 += 1;
172         }
173     }
174 }
175
```

```
Open 230104004174.c ~/Desktop
174 }
175
176 int successRate = trueCountFor2 * 100 / numberOfDataPoints;
177 fclose(dataFile3);
178
179 /* QUESTION -3 and MENU */
180 FILE *dataFile4;
181 dataFile4 = fopen("//home//gorkem//Desktop//test-final.txt" , "r");
182 if (dataFile4 == NULL) {
183     printf("An error occured while opening the file. \n");
184     return 1;
185 }
186
187 int choice;
188 int continueToInput = 1;
189 printf("MENU:\n 1.Explore Data(REsULT OF QUESTION-1).\n 2.Train Model(REsULT OF QUESTION-2).\n 3.Test Model(REsULT OF QUESTION-3).\n 4.EXIT.\n");
190
191 while(continueToInput == 1){
192     printf("Please choose the process: ");
193     scanf("%d", &choice);
194     switch(choice){
195     case 1:
196         printf("\nMinimum and Maximum X and Y values:\n");
197         printf("Class 0: X in [%d %d], Y in [%d %d] \n", min_x_0, max_x_0, min_y_0, max_y_0 );
198         printf("Class 1: X in [%d %d], Y in [%d %d] \n", min_x_1, max_x_1, min_y_1, max_y_1 );
199         break;
200
201     case 2:
202         printf("\nBest separator coordinates: X = %d, Y = %d, Total Error = %d \n", bestSeparatorX, bestSeparatorY, minError );
203         printf("Region Labels: \n");
204         printf("R1: Class Label %d \nR2: Class Label %d \nR3: Class Label %d \nR4: Class Label %d \n", bestR1, bestR2, bestR3, bestR4);
205         printf("success rate: %%%d \n \n", successRate );
206         break;
207
208     case 3:
209
210         while (fscanf(dataFile4, "%d %d", &sayi1, &sayi2) != EOF) { /* guesses for test-final text's points. according to best separators.*/
211             numberOfDataPoints += 1;
212             if (sayi1 <= bestSeparatorX && sayi2 > bestSeparatorY) {
213                 printf("For this point: X = %d, Y = %d ", sayi1, sayi2);
214                 printf("Class should be: %d \n", bestR1);
215             }
216             else if (sayi1 > bestSeparatorX && sayi2 > bestSeparatorY) {
217                 printf("For this point: X = %d, Y = %d ", sayi1, sayi2);
218                 printf("Class should be: %d \n", bestR2);
219             }
220             else if (sayi1 <= bestSeparatorX && sayi2 <= bestSeparatorY) {

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Open

230104004174.c
~/Desktop

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201     case 2:
202         printf("\nBest separator coordinates: X = %d, Y = %d, Total Error = %d\n",bestSeparatorX , bestSeparatorY ,mInError );
203         printf("Region Labels: \n");
204         printf("R1: Class Label %d \nR2: Class Label %d \nR3: Class Label %d \nR4: Class Label %d\n",bestR1,bestR2,bestR3,bestR4);
205         printf("success rate : %% %d \n\n" , successRate );
206         break;
207
208     case 3:
209
210
211         while (fscanf(dataFile4, "%d %d", &sayi1, &sayi2) != EOF){ /* guesses for test-final text's points. according to best separators.*/
212             numberOfDataPoints += 1;
213             if (sayi1 <= bestSeparatorX && sayi2 > bestSeparatorY) {
214                 printf("For this point: X = %d, Y = %d\n", sayi1 , sayi2);
215                 printf("Class should be: %d\n", bestR1);
216
217             }else if (sayi1 > bestSeparatorX && sayi2 > bestSeparatorY) {
218                 printf("For this point: X = %d, Y = %d\n", sayi1 , sayi2);
219                 printf("Class should be: %d\n", bestR2);
220             }else if (sayi1 <= bestSeparatorX && sayi2 <= bestSeparatorY) {
221                 printf("For this point: X = %d, Y = %d\n", sayi1 , sayi2);
222                 printf("Class should be: %d\n", bestR3);
223             }else if (sayi1 > bestSeparatorX && sayi2 <= bestSeparatorY) {
224                 printf("For this point: X = %d, Y = %d\n", sayi1 , sayi2);
225                 printf("Class should be: %d\n", bestR4);
226             }
227         }
228         fclose(dataFile4);
229         break;
230
231     case 4:
232         printf("The program is closing...\n");|
233         continueToInput = 0;
234         break;
235     default:
236         printf("\nPlease choose a valid process!\n");
237     }
238     if(continueToInput == 1){ /* check if user want to continue or not.*/
239         printf("\nIf you want to continue please press 1. Otherwise press any key: ");
240         scanf("%d",&continueToInput);
241     }
242 }
243
244
245
246 return 0;
247 }

```

-----OUTPUTS PARTS-----

```
gorkem@gorkem:~/Desktop$ ./230104004174
MENU:
 1.Explore Data(RESULT OF QUESTION-1).
 2.Train Model(RESULT OF QUESTION-2).
 3.Test Model(RESULT OF QUESTION-3).
 4.EXIT.
Please choose the process: 5

Please choose a valid process!

If you want to continue please press 1. Otherwise press any key: 0
gorkem@gorkem:~/Desktop$ ./230104004174
MENU:
 1.Explore Data(RESULT OF QUESTION-1).
 2.Train Model(RESULT OF QUESTION-2).
 3.Test Model(RESULT OF QUESTION-3).
 4.EXIT.
Please choose the process: 1

Minimum and Maximum X and Y values:
Class 0: X in [1 48], Y in [3 45]
Class 1: X in [5 49], Y in [0 49]

If you want to continue please press 1. Otherwise press any key: 1
Please choose the process: 2

Best separator coordinates: X = 15, Y = 27, Total Error = 14
Region Labels:
R1: Class Label 0
R2: Class Label 1
R3: Class Label 1
R4: Class Label 0
success rate : % 83

If you want to continue please press 1. Otherwise press any key: 0
gorkem@gorkem:~/Desktop$
```

```
Best separator coordinates: X = 15, Y = 27, Total Error = 14
Region Labels:
R1: Class Label 0
R2: Class Label 1
R3: Class Label 1
R4: Class Label 0
success rate : % 83
```

If you want to continue please press 1. Otherwise press any key: 0

```
gorkem@gorkem:~/Desktop$ ./230104004174
```

MENU:

- 1.Explore Data(RESULT OF QUESTION-1).
- 2.Train Model(RESULT OF QUESTION-2).
- 3.Test Model(RESULT OF QUESTION-3).
- 4.EXIT.

Please choose the process: 2

```
Best separator coordinates: X = 15, Y = 27, Total Error = 14
Region Labels:
R1: Class Label 0
R2: Class Label 1
R3: Class Label 1
R4: Class Label 0
success rate : % 83
```

If you want to continue please press 1. Otherwise press any key: 1

Please choose the process: 3

```
For this point: X = 26, Y = 31 Class should be: 1
For this point: X = 27, Y = 15 Class should be: 0
For this point: X = 22, Y = 48 Class should be: 1
For this point: X = 13, Y = 27 Class should be: 1
For this point: X = 27, Y = 27 Class should be: 0
For this point: X = 25, Y = 32 Class should be: 1
For this point: X = 3, Y = 30 Class should be: 0
For this point: X = 5, Y = 8 Class should be: 1
For this point: X = 41, Y = 45 Class should be: 1
For this point: X = 23, Y = 29 Class should be: 1
For this point: X = 29, Y = 12 Class should be: 0
For this point: X = 10, Y = 43 Class should be: 0
For this point: X = 45, Y = 5 Class should be: 0
```

If you want to continue please press 1. Otherwise press any key: 1

Please choose the process: 4

The program is closing...

```
gorkem@gorkem:~/Desktop$
```

```
gorkem@gorkem:~/Desktop$ ./230104004174
MENU:
1.Explore Data(RESULT OF QUESTION-1).
2.Train Model(RESULT OF QUESTION-2).
3.Test Model(RESULT OF QUESTION-3).
4.EXIT.
Please choose the process: 4
The program is closing...
gorkem@gorkem:~/Desktop$
```

```
gorkem@gorkem:~/Desktop$ ./230104004174
MENU:
1.Explore Data(RESULT OF QUESTION-1).
2.Train Model(RESULT OF QUESTION-2).
3.Test Model(RESULT OF QUESTION-3).
4.EXIT.
Please choose the process: 3
For this point: X = 26, Y = 31 Class should be: 1
For this point: X = 27, Y = 15 Class should be: 0
For this point: X = 22, Y = 48 Class should be: 1
For this point: X = 13, Y = 27 Class should be: 1
For this point: X = 27, Y = 27 Class should be: 0
For this point: X = 25, Y = 32 Class should be: 1
For this point: X = 3, Y = 30 Class should be: 0
For this point: X = 5, Y = 8 Class should be: 1
For this point: X = 41, Y = 45 Class should be: 1
For this point: X = 23, Y = 29 Class should be: 1
For this point: X = 29, Y = 12 Class should be: 0
For this point: X = 10, Y = 43 Class should be: 0
For this point: X = 45, Y = 5 Class should be: 0

If you want to continue please press 1. Otherwise press any key: 1
Please choose the process: 2

Best seperator coordinates: X = 15, Y = 27, Total Error = 14
Region Labels:
R1: Class Label 0
R2: Class Label 1
R3: Class Label 1
R4: Class Label 0
success rate : % 83

If you want to continue please press 1. Otherwise press any key: 6
gorkem@gorkem:~/Desktop$
```

YOUTUBE LINK : <https://youtu.be/XJ5cYKtXzXM>